Mega XL — custom upgrades & build notes

Upgrades to a Kickstarter v1.0 Mega XL CNC router.

This documents the changes made to a first-generation (Kickstarter v1.0) Mega XL: a taller, stiffer Z, closed-loop motors, all-proximity homing, and a complete electronics rebuild around a grblHAL controller in a custom enclosure with a proper 48 V power system, air assist, and probing.

At a glance

Subsystem	Stock (v1.0)	Upgraded
Vertical clearance	stock	+3″ via Mark Smith riser kit
Z assembly	stock Z	Mark Smith Z gantry (only V-wheels + X motor mount reused)
Spindle	DeWalt DWP611 (1¼ HP router)	2.2 kW (3 HP) Φ80 mm water-cooled, ER20, 24k RPM, VFD-driven
Stepper motors	generic open-loop steppers	all-in-one closed-loop steppers (driver integrated in motor)
Limit / homing	mechanical limit switches	proximity sensors on every axis
Probing	—	touch plate + 3D probe in parallel; grounded spindle (no tool clip)
Controller	8-bit Arduino board	grblHAL Teensy 4.x (USB + Ethernet)
Electronics enclosure	cramped stock steel box	custom 12×16×6″ aluminum / plexiglass
Power	stock	48 V Meanwell system, E-stop, buck-derived 12 V / 5 V rails
Air assist	none	Mist-relay 12 V solenoid → 30 PSI nozzle at the cut
Drag chains	stock 1.5	Y axis upsized to 2″ to fit the full cable bundle
Dust collection	—	4″ swivel + 5′ flex to a spindle dust shoe, on an overhead reel

Frame & mechanical

- **Mark Smith riser kit** — adds **3″ of additional vertical (Z) clearance**, allowing taller stock and longer tooling.
- **Mark Smith Z gantry** — a **complete replacement of the Z assembly**. The only stock parts carried over are the **four V-wheels** and the **X stepper motor mount**; everything else is new.

Spindle & cooling

The stock **DeWalt DWP611** (1¼ HP trim router) was replaced with a **2.2 kW (3 HP) water-cooled spindle** kit:

- **Spindle** — Φ80 mm body, 220 V, water-cooled, **ER20 collet**, up to **24,000 RPM at 400 Hz**.
- **VFD** — 2.2 kW (3 HP) inverter, fed from the 220 V circuit through the E-stop and commanded by the breakout board's **0–10 V PWM** speed signal (see Power distribution).
- **Mount** — 80 mm spindle clamp on the new Z gantry.
- **Braking resistor** — a 150 W resistor on the VFD dissipates regen energy for a fast stop (spindle down in **under 6 s**).

Coolant loop. A pump **submerged in a 1-gallon reservoir** circulates coolant through the spindle via **two 8 mm tubes** (supply and return): *reservoir → pump → spindle → radiator → reservoir*. The radiator is cooled by the 12 V fan (see the power tree). The pump is **interlocked to the spindle** — a **VFD digital output** drives the **NO contact of a 110 V relay** the pump plugs into — so it runs whenever the spindle does and the spindle can't be run dry. (Separate from the breakout board's Mist relay, which handles air assist.) The coolant tubes, the air-assist line, and dust extraction all reach the spindle through the overhead umbilical (below).

Motors

The generic steppers that shipped with the Mega XL were replaced with **all-in-one closed-loop stepper motors** — each motor has its stepper driver built into the motor body. This removes the separate driver boards, adds closed-loop position feedback (lost-step detection / alarm), and simplifies wiring to a single cable per motor.

- **Individual enable** per motor — each stepper's `EN` line is driven separately.
- The motors' `AL` **(alarm) outputs are wire-OR'd together** and fed to the controller's `ST0` **input**, so a stall / lost-step fault on any motor signals the controller.

All four axes are closed-loop. The machine is a **dual-Y gantry** — the two Y gantries are driven by ganged motors (**Y + A**), one per side.

Axis	Holding torque	Drives
Y + A	3.2 N·m each	the two Y gantries (ganged)
X	2.0 N·m	spindle carriage along the cross-beam (rack & pinion)
Z	1.2 N·m	Z

All motors have an **8 mm output shaft with a flat**, with pinions/couplings retained by a grub screw on the flat. *Build note:* the stock Mega XL pinions were 6 mm-bore and were **reamed to 8 mm** at a machine shop — this thinned the set-screw boss and reduced grub-screw thread engagement. See the rack-and-pinion maintenance appendix.

Homing & limit sensing

The Mark Smith Z gantry uses a **proximity sensor** as its limit switch. To keep one sensing scheme across the machine, **all remaining stock mechanical limit switches were replaced with the same style of proximity sensor**, so every axis homes on identical, non-contact hardware.

Probing

- The **touch-plate input is wired in parallel with the primary probe input**, alongside the **3D mechanical probe** — either device can trigger the same probe input.
- The **spindle mount is grounded**, so the tool itself is at ground through the spindle. That completes the touch-plate circuit on contact — **no alligator clip on the tool** is required.

Controller & electronics enclosure

- Discarded the **cramped stock steel electronics enclosure** and the **stock 8-bit Arduino controller board**.
- Control is now a **grblHAL Teensy 4.x** controller on a breakout board (32-bit).
- Built a **custom 12″ × 16″ × 6″ enclosure**:
  - Frame: **2020 aluminum extrusion**.
  - Five sides: **0.0625″ (1/16″) aluminum sheet**.
  - Top: **1/8″ clear plexiglass** (visibility into the electronics).

Connectivity

The Teensy 4.x supports Ethernet, and the breakout board has the **Ethernet option installed**. Both links to the Teensy are brought out as **panel patch-through connectors on the side of the enclosure**: a **female USB-A** jack and a **female Ethernet (RJ45)** jack.

Power distribution

A single **220 V, 20 A branch circuit** feeds the machine through a **front-panel emergency-stop button** (the E-stop breaks the 220 V input, so it drops both the PSU and the spindle drive). The 220 V powers an **800 W 48 V Meanwell power supply** and the **VFD** (spindle drive). Everything else is derived from the 48 V rail through two buck converters.

220 V, 20 A → front-panel E-stop →

  └ 48 V Meanwell power supply

  └ VFD (spindle drive)

48 V (PSU output) →

  └ 4 × stepper motors

  └ 12 V / 3 A buck converter

  └ 5 V / 3 A buck converter

12 V (buck) →

  └ 4 × proximity limit sensors

  └ 12 V fan cooling the spindle coolant loop (runs while the spindle runs)

  └ grblHAL Teensy 4.x breakout board, 12 V input — the board uses 12 V to generate the **0–10 V PWM spindle-speed signal** to the VFD

5 V (buck) →

  └ breakout board 5 V input — board logic power, **separate from the Teensy** (the Teensy is powered over USB)

  └ 3D probe

Relay outputs & air assist

The breakout board provides **5 relay outputs**, each **jumper-selectable for 5 V or 12 V**. Only the **Mist** output is in use, driving the air-assist nozzle:

> Mist relay → 12 V **NC** solenoid valve → regulator @ 30 PSI → 1/4″ tube (up the umbilical) → needle valve at top of Z gantry → 1/8″ copper airline → swivel nozzle aimed where the tool meets the stock

Connectors & terminations

Ferrules are crimped on all terminal-block connections — every stranded conductor landing on a terminal block is ferruled for a clean, captured, reliable termination.

Enclosure pass-throughs use **Molex Mini-Fit Jr panel-mount connectors**. The choice was deliberate: with roughly 60 terminations to make, the builder had far more confidence producing **~60 quality crimps** than ~60 quality solder joints on aircraft-style connectors.

Cable & wiring schedule

All runs use shielded cable.

Cable	Run	Conductors
18/6 shielded	enclosure → each stepper motor	6 (assignment depends on motor type — see below)
22/3 shielded	enclosure → Y-axis proximity switches	`12V`, `Sig`, `GND`
22/6 shielded	enclosure → X & Z proximity sensors + 3D probe (shared)	`12V`, `GND`, `XLim`, `ZLim`, `P`, `5V`

18/6 stepper cable — conductor assignment

Conductor	Closed-loop stepper	Discrete (open-loop) stepper
1	`48V`	`A+`
2	`GND`	`A-`
3	`DIR`	`B+`
4	`PUL`	`B-`
5	`EN`	`EN`
6	`AL`	`AL`

All four axes on this build are closed-loop, so they all use the left-column assignment (`48V` / `GND` / `DIR` / `PUL` / `EN` / `AL`). `DIR` / `PUL` are direction/step to the integrated drives; `EN` = enable, `AL` = alarm. The discrete `A±` / `B±` column is kept for reference only — the same 18/6 cable would serve an open-loop motor if one were ever fitted.

Drag chains (cable carriers)

Each axis's drag chain carries every cable that has to continue past it, so loading is cumulative toward the base of the motion chain. **Y is the fullest** — everything routes through the Y carrier first to reach the gantry, then the X carrier feeds the spindle carriage, while the opposite gantry rail (**A**) carries only its own stepper. The two limit-sensor drops that don't need to flex (A and Y) **begin after** their drag chains rather than riding through them.

Drag chain	Cables carried
A axis (opposite gantry rail)	1 × 18/6 (A stepper) *A limit (22/3) picks up after the chain*
X axis (spindle carriage)	1 × 12/4 (spindle power) 2 × 18/6 (X & Z steppers) 1 × 22/6 (X & Z limits + 3D probe)
Y axis (gantry — carries everything)	1 × 12/4 (spindle power) 3 × 18/6 (X, Z & Y steppers) 1 × 22/6 (X & Z limits + 3D probe) *Y limit (22/3) picks up after the chain*

That Y bundle — one 12/4, three 18/6, and one 22/6 — would not fit the stock **1.5** drag chain, so the Y axis was moved up to the **next size larger (2”)** carrier.

Overhead umbilical & dust collection

The spindle's services arrive from **above** rather than through a drag chain, on a **ceiling-mounted spring retraction reel** that takes up slack as the gantry moves.

- **Umbilical:** the two 8 mm coolant tubes and a 1/4″ compressed-air tube run from the top of the Z gantry into a **1” cable sleeve**.
- **Retraction reel:** the sleeve hangs from a spring-loaded retraction reel on the ceiling above the machine. A **3D-printed cradle** holds the sleeve and clips to the reel hook, so the reel's pull-wire can't wrap and choke the sleeve while still allowing free movement.
- **Routing:** the sleeve terminates at the **back of the CNC**, where the coolant tubes join the pump outlet and the radiator inlet (radiator outlet returns to the reservoir). The air tube comes from the

pressure-regulator outlet, whose inlet is fed by the 12 V NC Mist solenoid.

- **Dust collection:** alongside the umbilical, a **4″ duct** ends in a **4″ swivel** feeding **5′ of 4″ flex hose** to a **dust shoe** on the spindle.

Rack-and-pinion drive — setup & maintenance

X (and the Y/A gantry drives) run on rack and pinion. After ~10 hours the X pinion crept **left** under load. A closed-loop motor can't see slip *downstream* of its shaft encoder, so a pinion loosening on the shaft is invisible until a fixed reference (the toolsetter at G59.3) exposes it. With the bores reamed 6→8 mm the set-screw boss is thin and grub-screw grip is the weak point; X slips first because it takes the most direct cutting side-load, but the other reamed pinions carry the same risk.

Pinion-to-shaft — stop the slip

1. Pull the pinion; confirm the grub screw seats on the shaft **flat** (not a round section), and check whether the screw marred the flat or only kissed it.
2. **Degrease** shaft, bore, and screw threads — oil defeats both grip and threadlocker.
3. Use the **longest grub screw** the thinned boss will take, on the flat; add a second screw if there's room.
4. **Threadlocker:** blue (242/243) is normally enough; with the reduced thread engagement here, **red (271)** is justified as insurance against back-out — just note removal then needs heat (~250 °C, easy on a bare pinion).
5. Torque to screw size (M3 ~1.0–1.4, M4 ~2.5–3, M5 ~5–6 N·m); snug, don't strip the hex. Let red cure ~24 h before heavy cuts.

Pinion-to-rack mesh & backlash

1. Loosen the motor mount so the pinion can move toward / away from the rack.
2. Shim **0.05–0.10 mm** (feeler gauge, or ~0.1 mm folded paper) between a pinion tooth and the rack, push the pinion in against the shim, tighten the mount, remove the shim.
3. Check the mesh over the **full travel** — it should stay lightly meshed everywhere. Tight at one end + slack at the other means the rack isn't parallel to the axis; realign the rack first.
4. Confirm full tooth-width engagement (pinion not riding a tooth edge).

Verify

1. By hand: smooth full travel, minimal lash; rock the gantry and feel for play at the pinion.
2. Dial indicator: commanded vs actual on small moves; reverse direction to read backlash (a few thou).

3. Re-run a heavy back-and-forth / cut loop with paint-pen **witness marks** (motor shaft ↔ pinion, and a pinion tooth ↔ a rack tooth) — no separation = fixed.
4. Confirm the `AL → ST0` alarm actually halts grblHAL (force a stall) so a real fault can't pass silently.

If it still creeps

A single grub screw — especially in a reamed, thin-wall boss — is marginal for rack-and-pinion loads. Durable fixes, in order: **proper 8 mm-bore pinions** (full set-screw boss) → **clamp / split-hub pinion** (grips the whole shaft, no set screw) → **keyed shaft**. Given the reamed pinions, the **clamp-hub style is the recommended upgrade** — it ends set-screw slip for good.

Sourcing a replacement pinion

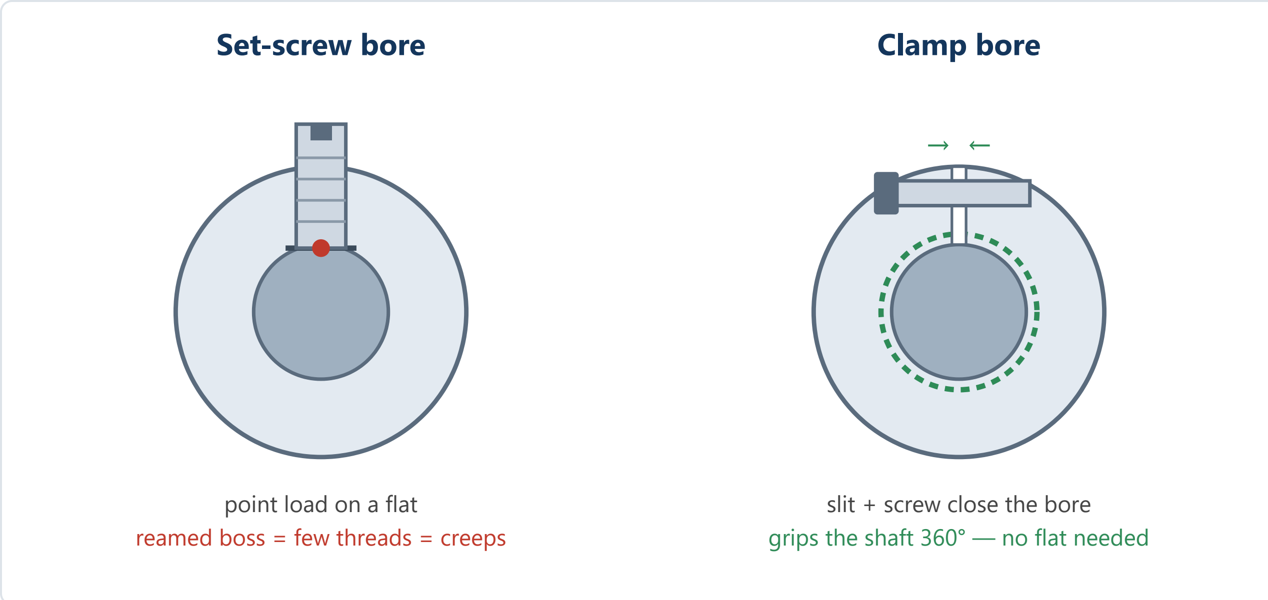
Match the rack on three things, plus the bore:

- **Module** (tooth size) — `module = tooth pitch ÷ π` (pitch 3.14 mm = Mod 1, 4.71 mm = Mod 1.5, 6.28 mm = Mod 2). Must match the rack exactly.
- **Pressure angle** — almost always **20°**.
- **Tooth count** — match the original to keep steps/mm, or pick a new count and recalibrate `$100` (X steps/mm) in grblHAL.
- **Bore** — 8 mm, ideally a **clamp or keyed** bore, not a bare set-screw boss.

Sources that let you specify *module + 20° + 8 mm clamp/keyed bore*: **KHK / QTC** (khkgears.us, qtcgears.com — best for CNC rack & pinion), **SDP/SI** (sdp-si.com), **Misumi** (configurable bore & keyway), and **McMaster-Carr** (quick metric-module steel gears). Budget CNC rack-and-pinion sets are on Amazon / AliExpress (Mod 1/1.5, 20T, often 8 mm bore) with variable quality.

Clamp hub vs set-screw hub

A **clamp (split) hub** grips like a clamp-style shaft collar: a slit is sawn through the bore and a screw bridges it; tightening the screw squeezes the slit closed so the **whole bore clamps the shaft 360°** — far more holding torque, no marring, and no dependence on a thin set-screw boss (exactly the reamed-pinion weakness).



Two routes to a clamp mount: a pinion with an **integral clamp bore** (specify it when ordering from KHK / Misumi / SDP), or a **keyless locking bushing** (Trantorque GT, Fenner B-LOC, …) dropped inside a larger-bore pinion — the bushing expands to grip both the shaft and the bore, letting you use almost any pinion of the right module on the 8 mm shaft. The latter is handy here, since matching bore *and* gearing was the sticking point.

Bill of materials

Group	Item	Spec / source
Machine	Mega V XL (Kickstarter edition)	base machine
Frame	Mark Smith riser kit	+3″ Z clearance
Frame	Mark Smith Z gantry assembly	full Z replacement
Control	Teensy 4.1 + Teensy 4.x breakout board	grblHAL, Ethernet option
I/O	USB-A pass-through, panel mount	
I/O	Ethernet pass-through, panel mount	
I/O	E-stop button	breaks 220 V input
Power	Meanwell 48 V power supply	800 W
Power	C20 panel-mount inlet	220 V mains in
Power	Power relay accessory	110 V pump relay — Amazon B07Z6R1KP3
Spindle	2.2 kW 80 mm water-cooled spindle + 2.2 kW VFD kit	ER20, 24k RPM — Amazon B073TRP92V
Spindle	Braking resistor	150 W (<6 s stop)
Cooling	Submersible pump	110 V
Cooling	Coolant reservoir	1 gal container (holds the pump)
Cooling	Radiator with integrated 12 V fan(s)	
Cooling	8 mm clear tubing	25′ (coolant lines)
Cooling	Coolant mix	1 qt antifreeze + ½ gal distilled water + a few drops of bleach
Motion	Z stepper (closed-loop)	1.2 N·m
Motion	X stepper (closed-loop)	2.0 N·m
Motion	Y + A steppers (closed-loop)	3.2 N·m each (×2)
Motion	Drag chain	2″ (Y-axis upsize)
Sensing	Proximity sensors	×4 (limits / homing)
Sensing	3D mechanical probe	

Group	Item	Spec / source
Sensing	Toolsetter probe	
Air assist	1/4″ black tubing	10′
Air assist	Needle valve	
Air assist	Copper tubing	12″
Air assist	Swivel nozzle	1/8″
Air assist	Solenoid valve	12 V, NC
Air assist	Pressure regulator	set 30 PSI
Air assist	3D-printed nozzle bracket	bolts to Z; aims the air stream at the tool tip
Cabling	Shielded cable	22/6, 22/3, 18/6

Not yet itemized (described in the build above but missing from this BOM):

- **Molex Mini-Fit Jr connectors** — panel-mount + inline housings, pins and crimp terminals (count per the wiring schedule) — *still to be listed*.
- **Enclosure** — 2020 extrusion, 1/16″ aluminum sheet (5 sides), 1/8″ plexiglass top.
- **Buck converters** — 12 V 3 A and 5 V 3 A.
- **Ferrules** for the terminal-block connections.
- **Touch plate** (probe input in parallel with the 3D probe).
- **Dust collection** — 4″ duct, 4″ swivel, 5′ of 4″ flex hose, dust shoe.
- **Overhead service rig** — spring retraction reel, 1″ cable sleeve, 3D-printed sleeve cradle.